STT 861: Theory of Probability & Statistics I

Lecture 17: Continuous Distributions I

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Oct 22, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 3.3

Where we are: from counts to measurements

Done (§3.1–3.2): discrete families

- Bernoulli, Binomial, Hypergeometric
- Poisson, Geometric, Negative Binomial

Today (§3.3): continuous families

- Uniform, Gamma (Exponential, χ^2, Weibull)
- Normal, Beta
- Cauchy, Lognormal, Double exponential

Same checklist as before

For each family: **story**, pdf, support, parameters, $\mathbb{E}X$, $\mathrm{Var}\,X$, mgf, relationships.

Roadmap

Family	Support	Typical story
Uniform	$[a, b]$	"no preference"
Exponential	$(0, \infty)$	time to next event
Gamma	$(0, \infty)$	time to α-th event
χ^2_p	$(0, \infty)$	sum of squared normals
Normal	$\mathbb{R}$	errors, sums, averages
Beta	$(0, 1)$	a proportion
Cauchy	$\mathbb{R}$	ratio of normals; heavy tails
Lognormal	$(0, \infty)$	multiplicative growth
Double exp.	$\mathbb{R}$	sharp peak, fat-ish tails

Motivation: questions that need a continuous model

Question	Natural model
A bus comes "sometime in the next 20 minutes" — how long do I wait?	Uniform
How long until the next customer arrives? Until the 5th?	Exponential / Gamma
How accurate is a measurement; what is the spread of heights?	Normal
What fraction of voters / defective parts / conversions?	Beta
Why does the average of Cauchy data *not* settle down?	Cauchy
Why are incomes and stock prices skewed?	Lognormal
Noise with a sharp peak and fatter tails than normal?	Double exponential

Think (1 min)

A continuous X has $P(X = x) = 0$ for every x. So how do we describe its distribution, and what does a "family" give us? *(Answer: pdf f with $P(a < X < b) = \int_a^b f$; the family gives a few parameters that determine f.)*

Uniform(a, b): "all values equally likely"

Definition

$X \sim \text{Uniform}(a, b)$, $a < b$:

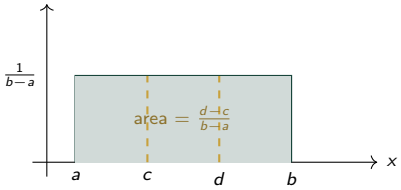
$$f(x \mid a, b) = \frac{1}{b-a}, \quad a \le x \le b \text{ (0 else)},$$

$$F(x) = \frac{x-a}{b-a} \ (a \le x \le b).$$

Mean and variance

$$\mathbb{E}X = \frac{a+b}{2}, \qquad \text{Var}\, X = \frac{(b-a)^2}{12}.$$

Proof. $\mathbb{E}X = \int_a^b \frac{x}{b-a} dx = \frac{b^2-a^2}{2(b-a)}$, and $\mathbb{E}X^2 = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}$; subtract $\left(\frac{a+b}{2}\right)^2$.



Probability = (length of interval)/(length of support)

Why it matters

Uniform(0, 1) is the raw material of simulation: every other distribution can be generated from it (next slide).

Example (bus) and a key use: the probability integral transform

Example: waiting for a bus

A bus arrives at a uniformly random time in the next 20 minutes: $X \sim \mathrm{Uniform}(0, 20)$.

(a) $P(X > 15) = \dfrac{20 - 15}{20} = \dfrac{1}{4}$.

(b) $\mathbb{E}X = 10$ min, $\mathrm{Var}\, X = \dfrac{20^2}{12} = \dfrac{100}{3}$.

(c) Given you already waited 8 min, $P(X > 15 \mid X > 8) = \dfrac{5}{12}$: still uniform on the rest.

Probability integral transform (Ch. 2)

If X has a continuous, strictly increasing cdf F, then $F(X) \sim \mathrm{Uniform}(0, 1)$. Conversely $F^{-1}(U) \sim F$ for $U \sim \mathrm{Uniform}(0, 1)$.

Simulating an exponential

$F(x) = 1 - e^{-x/\beta}$ so $F^{-1}(u) = -\beta \log(1 - u)$. Draw U, set $X = -\beta \log U$ (same law as $-\beta \log(1 - U)$): $X \sim$ Exponential(β).

Gamma: motivation — the time of the α-th event

Story

Customers arrive at a help desk as a Poisson process with rate λ per hour. Let T_k be the time of the k-th arrival.

- T_1 is the wait for the first arrival.
- T_k is the wait for the k-th.

Both are positive and continuous. The family that fits is the **gamma** (with $\alpha = k$, $\beta = 1/\lambda$).

The gamma function

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\, dt, \quad \alpha > 0.$$

- $\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$ (integration by parts)
- $\Gamma(n) = (n-1)!$ for integers $n \geq 1$
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

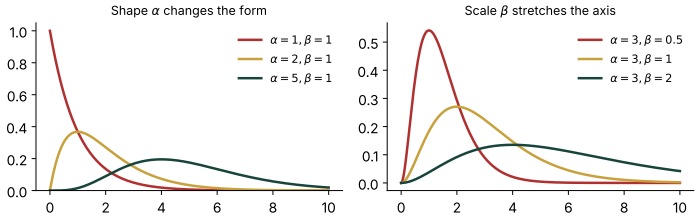
Check

$\Gamma(\frac{5}{2}) = \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$.

Gamma(α, β): definition, shape and moments

Definition (shape $\alpha > 0$, scale $\beta > 0$)

$$f(x \mid \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^{\alpha}} x^{\alpha-1} e^{-x/\beta}, \qquad x > 0.$$



Moments and mgf

$\mathbb{E}X = \alpha\beta, \quad \operatorname{Var} X = \alpha\beta^2, \quad M_X(t) = (1 - \beta t)^{-\alpha}, \ t < 1/\beta.$

Gamma: deriving the mean and the mgf

Key idea: the kernel integrates to a Gamma function

Since $\int_0^\infty x^{\alpha-1}e^{-x/\beta}dx = \Gamma(\alpha)\beta^\alpha$,

$$\mathbb{E}X^n = \frac{1}{\Gamma(\alpha)\beta^\alpha}\int_0^\infty x^{\alpha+n-1}e^{-x/\beta}dx = \frac{\Gamma(\alpha+n)\beta^n}{\Gamma(\alpha)}.$$

- $n=1$: $\mathbb{E}X = \alpha\beta$.
- $n=2$: $\mathbb{E}X^2 = \alpha(\alpha+1)\beta^2$, so $\mathrm{Var}\,X = \alpha\beta^2$.

mgf

$$M_X(t) = \frac{1}{\Gamma(\alpha)\beta^\alpha}\int_0^\infty x^{\alpha-1}e^{-x(\frac{1}{\beta}-t)}dx.$$

This is the same kernel with scale $\beta' = \frac{\beta}{1-\beta t}$ (valid if $t < 1/\beta$), so

$$M_X(t) = \frac{\Gamma(\alpha)\beta'^\alpha}{\Gamma(\alpha)\beta^\alpha} = (1-\beta t)^{-\alpha}.$$

Consequence

Independent $X_i \sim \mathrm{Gamma}(\alpha_i, \beta) \Rightarrow \sum X_i \sim \mathrm{Gamma}(\sum \alpha_i, \beta)$ (multiply mgfs).

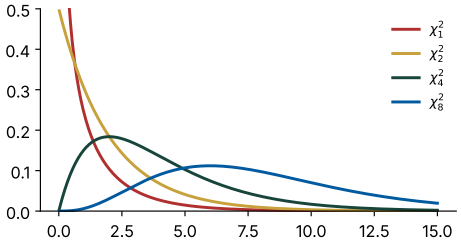
Special cases of the gamma

Three important sub-families

Name	= Gamma	Moments
Exponential(β)	$\alpha = 1$	$\mathbb{E}X = \beta,\ \mathrm{Var}\, X = \beta^2$
χ^2_p	$\alpha = \frac{p}{2}, \beta = 2$	$\mathbb{E}X = p,\ \mathrm{Var}\, X = 2p$
Weibull(γ, β)	$X^{1/\gamma}$ of Exp(β)	$\mathbb{E}Y = \beta^{1/\gamma}\Gamma(1+\frac{1}{\gamma})$

Exponential: $f(x) = \frac{1}{\beta}e^{-x/\beta}$, $F(x) = 1 - e^{-x/\beta}$

The $\alpha = 1$ gamma: the waiting time for **one** Poisson event, with $\beta = 1/\lambda$.



χ^2_p densities for $p = 1, 2, 4, 8$: the degrees of freedom p is a shape parameter.

Where does χ^2 come from? Squaring a standard normal

Fact

If $Z \sim N(0,1)$ then $Z^2 \sim \chi^2_1 = \mathrm{Gamma}(\frac{1}{2}, 2)$.

Derivation (transformation, Ch. 2, not monotone)

$Y = Z^2$. For $y > 0$: $F_Y(y) = P(-\sqrt{y} \le Z \le \sqrt{y}) = 2\Phi(\sqrt{y}) - 1$. Differentiating,

$$f_Y(y) = 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}}\, y^{-1/2} e^{-y/2} = \frac{1}{\Gamma(\frac{1}{2}) 2^{1/2}}\, y^{\frac{1}{2}-1} e^{-y/2}.$$

This is Gamma$(\frac{1}{2}, 2)$ because $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Consequence

$Z_1, \ldots, Z_p$ iid $N(0,1) \Rightarrow \sum Z_i^2 \sim \chi^2_p$ (add gamma shapes). So $\mathbb{E} \sum Z_i^2 = p$, Var $= 2p$, and $P(\chi^2_5 > 11.07) = 0.05$.

Exponential: the memoryless property

Memorylessness

If $X \sim \text{Exponential}(\beta)$ then for all $s, t > 0$

$$P(X > s + t \,|\, X > s) = P(X > t).$$

Proof. $\dfrac{P(X > s+t)}{P(X > s)} = \dfrac{e^{-(s+t)/\beta}}{e^{-s/\beta}} = e^{-t/\beta}.$

Interpretation

A used part that has survived s hours is "as good as new": past survival carries no information about the future. The exponential is the only continuous distribution with this property.

Example: light bulb

Lifetime $X \sim$ Exponential with mean 1000 h.

(a) $P(X > 500) = e^{-1/2} \approx 0.607$.

(b) $P(X > 1500 \,|\, X > 500) = P(X > 1000) = e^{-1} \approx 0.368$.

(c) Median: $1 - e^{-m/1000} = \frac{1}{2} \Rightarrow m = 1000 \ln 2 \approx 693$ h, less than the mean.

Check

Is the discrete analogue memoryless? *(Yes: the Geometric, §3.2.)*

Gamma and Poisson are two views of one process

Gamma–Poisson relationship (integer α)

If $X \sim \text{Gamma}(\alpha, \beta)$ with α an integer and $Y \sim \text{Poisson}(x/\beta)$,

$$P(X \le x) = P(Y \ge \alpha).$$

Why

"The α-th event happens by time x" $\iff$ "at least α events occurred in $[0, x]$", and the count in time x is Poisson with mean x/β when events occur at rate $1/\beta$.

Example

Arrivals at rate 2 per hour. $T_3 \sim \text{Gamma}(3, \frac{1}{2})$. Find $P(T_3 > 2)$.

Number of arrivals in 2 h is $Y \sim \text{Poisson}(4)$:

$$P(T_3 > 2) = P(Y \le 2) = 13e^{-4} \approx 0.238,$$

since $e^{-4}(1 + 4 + \frac{4^2}{2}) = 13e^{-4}$. No integral needed.

Weibull: lifetimes with ageing

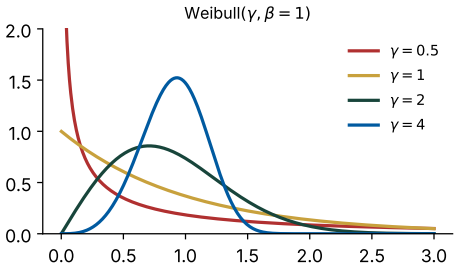
Weibull(γ, β), $\gamma, \beta > 0$

$$f(y) = \frac{\gamma}{\beta} y^{\gamma-1} e^{-y^\gamma/\beta}, \quad y > 0.$$

If $X \sim \text{Exponential}(\beta)$ then $Y = X^{1/\gamma} \sim \text{Weibull}(\gamma, \beta)$.

Derivation

$X = Y^\gamma$, $\frac{dx}{dy} = \gamma y^{\gamma-1}$, so $f_Y(y) = \frac{1}{\beta} e^{-y^\gamma/\beta} \cdot \gamma y^{\gamma-1}$ (monotone transformation, Thm 2.1.5).



- $\gamma = 1$: exponential (constant failure rate).
- $\gamma > 1$: failure rate increases (wear-out). $\gamma < 1$: decreases (infant mortality).

Activity

Help-desk arrivals: Poisson process, rate 6 per hour

(a) Name the distribution of the time T_1 (hours) until the first arrival. Find $P(T_1 > \frac{1}{4})$.

(b) Name the distribution of the time T_4 until the 4th arrival. Find $\mathbb{E}\,T_4$ and $\mathrm{Var}\,T_4$.

(c) Use the gamma–Poisson relationship to find $P(T_4 > 1)$.

(d) It is now 30 minutes in and nobody has come. What is the chance that still nobody comes in the next 15 minutes?

Hint: $\beta = 1/\lambda$ is the scale; "at least 4 events" $\Leftrightarrow$ $Y \geq 4$ for $Y \sim \mathrm{Poisson}(6)$.

Activity — Solutions

(a) $T_1 \sim \text{Exponential}(\beta = \frac{1}{6})$. $P(T_1 > \frac{1}{4}) = e^{-6/4} = e^{-1.5} \approx 0.223$.

(b) $T_4 \sim \text{Gamma}(4, \frac{1}{6})$: $\mathbb{E}\,T_4 = 4 \cdot \frac{1}{6} = \frac{2}{3}$ h (40 min), Var $T_4 = 4 \cdot \frac{1}{36} = \frac{1}{9}$.

(c) $Y \sim \text{Poisson}(6)$ counts arrivals in 1 h. $P(T_4 > 1) = P(Y \le 3) = e^{-6}(1 + 6 + 18 + 36) = 61e^{-6} \approx 0.151$.

(d) Memoryless: $P(T_1 > \frac{3}{4} \mid T_1 > \frac{1}{2}) = P(T_1 > \frac{1}{4}) = e^{-1.5} \approx 0.223$.